

Minimal left ideals are not matrix factors

Conjecture 00000007627

earthking11

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Statement and interpretation

The source defines ideals as left-module structures and asserts that their minimal-ideal count equals the matrix-factor count, with a center-dimension counting law. It gives no ring or field restriction. We use the usual meaning of minimal nonzero left ideals and matrix factors in a finite direct-product decomposition. We count distinct ideals, not their module isomorphism classes. The latter is a different assertion.

Theorem 1. *Let $K = \mathbb{F}_2$ and $R = M_2(K)$, considered as its own left regular module. The actual finite set of minimal left ideals has cardinality at least 2. Nevertheless R is a single simple matrix factor, and its center has dimension 1 over K . Thus the two claimed counts disagree.*

Proof. Consider the two column-supported ideals

$$L_0 = \left\{ \begin{pmatrix} a & 0 \\ b & 0 \end{pmatrix} : a, b \in K \right\}, \quad L_1 = \left\{ \begin{pmatrix} 0 & a \\ 0 & b \end{pmatrix} : a, b \in K \right\}.$$

They are closed under addition and left multiplication by arbitrary matrices in R . Each is nonzero, and E_{00} belongs to L_0 but not L_1 , so they are distinct.

To prove minimality, take a nonzero matrix $A \in L_0$. At least one entry in its first column is 1. Let its row index be j . For any $B \in L_0$, choose C whose column j is the first column of B , with its other column zero. Then $CA = B$. Hence any nonzero left ideal inside L_0 contains all of L_0 . This proves minimality. The same argument, using the second column, proves that L_1 is minimal.

Since R is finite, its set of submodules and the subset of minimal left ideals are finite. The two distinct ideals give an injection from a two-element set, so the actual minimal-left-ideal count is at least 2. On the other hand, R is a simple matrix ring and has the genuine one-factor decomposition

$$R \cong \prod_{i \in \{0\}} M_2(K), \quad A \mapsto (i \mapsto A).$$

The inverse evaluates at the sole index, and the actual factor-index set has cardinality 1. Finally, the center of $M_2(K)$ is the scalar copy of K , so its dimension over K is 1. Neither count can equal the minimal-left-ideal count. $\square$

Formal verification and limits

Lean uses the actual matrix ring, its left regular module `Submodule R R`, the subtype of all minimal left ideals, and the actual `Subalgebra.center`. It proves minimality against every nonzero subideal, finite cardinality, the injection, the one-factor ring equivalence, ring simplicity and the center dimension. The capstone combines the decomposition witness with both actual counting mismatches.

The project builds on Lean 4.33.1 with pinned Mathlib; printed audits use only propositional extensionality, classical choice and quotient soundness. No unfinished proof, native evaluation, extra axiom or long search is used. The source does not restrict this problem to Clifford algebras; no claim that this example is a Clifford algebra is needed or made. We do not assign a new meaning to its unspecified “direct-sum index.”